



Release Notes G3 v2.1.0

Summary

This document describes the target devices, supported tools and features of the G3 Software Stack release, with information regarding enhancements, bug fixes and known issues. This G3 v2.1.0 supports Hybrid Specification as defined by the G3 Alliance.

Table of Contents

Summary	1
Table of Contents	2
1 Description	3
1.1 Versions	3
1.2 Target Devices	3
1.3 Supported Boards	3
1.4 Supported Development Tools	4
1.5 Application Tools	4
2 Applicable Documents	5
3 Release Contents	5
3.1 Provided Example Applications:	5
3.2 PHY Examples support:	6
3.3 G3 Examples Support:	8
4 New Features / Changes	10
5 Resolved Issues	11
6 Known Issues	11
7 Memory Footprint	12
7.1 PL360/PL460 Binaries	12
7.2 G3 Stack and other modules	12
7.2.1 G3 Device with HW Crypto capabilities	12
7.2.2 G3 Device with SW Crypto libraries	12
7.2.3 G3 Coordinator with HW Crypto capabilities for Networks up to 500 Devices	12
7.2.4 G3 Coordinator with HW Crypto capabilities for Networks up to 2000 Devices	13
7.2.5 G3 Stack Serialized to be managed from an external Host MCU/MPU	13
8 Revision History	14

1 Description

1.1 Versions

- PL360 PHY CEN-A: v36.01.05
- PL360 PHY FCC: v36.02.05
- PL360 PHY ARIB: v36.03.05
- PL360 PHY CEN-B: v36.04.05

- RF215 PHY: v1.1.3

- ADP: v2.3.2
- LOADng: v2.3.2
- LBP: 4.0.1
- PLC MAC: v1.7.7
- PLC MAC RT: v2.0.8
- RF MAC: v1.1.8

1.2 Target Devices

- PIC32CXMTSH and PL460
- PIC32CXMTSH and PL460 and AT86RF215
- PIC32CXMTG and PL460
- PIC32CXMTG and PL460 and AT86RF215
- PIC32CXMTC and PL460
- PIC32CXMTC and PL460 and AT86RF215
- SAM4CMS16C with PL460
- SAM4CMS16C with PL360
- SAME70Q21 with PL460
- SAME70Q21 with PL360
- SAMG55J19 with PL460
- SAMG55J19 with PL360
- SAMG55J19 with PL460 and AT86RF215
- SAMG55J19 with PL360 and AT86RF215
- PL485
- PL485 and AT86RF215

1.3 Supported Boards

- PIC32CXMTSH-DB + PL460-EK
- PIC32CXMTSH-DB + PL460-EK + ATREB215
- PIC32CXMTG-EK + PL460-EK
- PIC32CXMTG-EK + PL460-EK + ATREB215
- PIC32CXMTC-DB + PL460-EK
- PIC32CXMTC-DB + PL460-EK + ATREB215
- PL360MB-EK

- SAM4CMS-DB + PL460-EK
- SAME70XPLAINED + PL460-EK
- PL360BN
- SAMG55XPLAINED + PL460-EK
- SAMG55XPLAINED + PL460-EK + ATREB215
- PL360G55CF-EK
- PL360G55CB-EK
- PL360G55CF-EK + ATREB215
- PL360G55CB-EK + ATREB215
- PL485-EK
- PL485-EK + ATREB215

1.4 Supported Development Tools

- IAR Embedded Workbench for ARM v9.20.4
- Keil µVision 5.25
- Microchip Studio 7.0
- J-32 Debug Probe
- Atmel-ICE
- Segger J-Link

1.5 Application Tools

- Microchip Hybrid Sniffer v2.0.4
- Microchip PLC PHY Tester Tool v3.1.3
- MCHP Python Tools:
 - mchp_plc_tools_adp_mac_serialized v2.0.14
 - mchp_plc_tools_phy_tester_public v2.1.3
 - mchp_rf_tools_phy_tester_public v1.0.1
- MCHP USI Host v2.4.0

2 Applicable Documents

- *G3-PLC Firmware Stack including Hybrid Profile User Guide (rev. B)*
- *G3-PLC Alliance - G3-PLC Specifications - CENELEC - ARIB - FCC - revision (2022)*
- *G.9903 : Narrowband orthogonal frequency division multiplexing power line communication transceivers for G3-PLC networks* [[G.9903](#)]
- *Recommendation G.9903 (2017) Amendment 1 (05/21)* [[G.9903 Amendment 1](#)]
- *Recommendation G.9903 (2017) Corrigendum 1 (03/23)* [[G.9903 Corrigendum1](#)]
- *Recommendation G.9903 (2017) Amendment 2 (03/23)* [[G.9903 Amendment 2](#)]

3 Release Contents

3.1 Provided Example Applications:

Example Application	Description
PHY_SNIFFER_TOOL	Monitors frames in the G3 network and then sends them via USI serialization to a PC tool. Includes Hybrid Sniffer on supported platforms
PHY_TESTER_TOOL	Allows to check the complete performance of the Microchip G3 PHY Layers on boards using a PC Tool. Includes Hybrid Phy Tester on supported platforms
PHY_TX_TEST_CONSOLE	Demonstrates the complete performance of the Microchip G3 PHY Layer avoiding timing limitations in the PC host.
PHY_PLC_AND_GO	A chat application running on top of PL360/PL460 PHY Layer.
MACRT_PLC_AND_GO	A chat application running on top of PL360/PL460 MAC RT Layer.
ADP_MAC_SERIALIZED_APP	The ADP and MAC serialization is an application example that brings access to the ADP, MAC and LBP API through a serial connection. Includes Hybrid Profile examples
DLMS_APP_COORD	Shows how the G3 API along with an IPv6 stack should be used from a typical DLMS application, client side. As a coordinator implements the ability to create a Network and accept Joining nodes. Includes Hybrid Profile examples
DLMS_APP_DEV	Shows how the G3 API along with an IPv6 stack should be used from a typical DLMS application, server side. Includes Hybrid Profile examples
METER / MODEM APP	Same as DLMS_APP_DEV but separated at IP level and communicating through a Serial connection. METER implements DLMS and emulated metrology part. MODEM implements IPv6 and G3 communication layers.
DEMO_METER_G3	Device implementing real metrology and G3 Hybrid communications. Available for PIC32CXMTSH-DB and PIC32CXMTC-DB

3.2 PHY Examples support:

TARGET BOARD (Project folder prefix)	PHY_SNIFFER_TOOL	PHY_TESTER_TOOL	PHY_TX_TEST _CONSOLE	PHY_PLC_AND _GO
PL360MB (sam4cms16c_atpl360mb)	CEN-A CEN-B FCC	CEN-A CEN-B FCC	CEN-A CEN-B FCC	CEN-A CEN-B FCC
PL360BN (same70q21_pl360bn)	N/A	CEN-A CEN-B FCC	CEN-A CEN-B FCC	CEN-A CEN-B FCC
PL360G55CB (samg55j19_pl360g55cb_ek)	CEN-B	CEN-B	CEN-B	CEN-B
PL360G55CF (samg55j19_pl360g55cf_ek)	CEN-A FCC	CEN-A FCC	CEN-A FCC	CEN-A FCC
PL485-EK (samg55j19_pl485_ek)	CEN-B	CEN-B	CEN-B	CEN-B
PIC32CXMTG-EK + PL460 (pic32cxmtg_ek_pl460)	CEN-A CEN-B FCC	CEN-A CEN-B FCC	CEN-A CEN-B FCC	CEN-A CEN-B FCC
PIC32CXMTSH-DB + PL460 (pic32cxmtsh_db_pl460)	N/A	CEN-A CEN-B FCC	CEN-A CEN-B FCC	CEN-A CEN-B FCC
PIC32CXMTC-DB + PL460 (pic32cxmtc_db_pl460)	N/A	CEN-A CEN-B FCC	CEN-A CEN-B FCC	CEN-A CEN-B FCC
SAM4CMS-DB + PL460 (sam4cms_db_pl460)	N/A	CEN-A CEN-B FCC	CEN-A CEN-B FCC	CEN-A CEN-B FCC
SAME70XPLAINED + PL460 (same70_xplained_pl460)	N/A	CEN-A CEN-B FCC	CEN-A CEN-B FCC	CEN-A CEN-B FCC
SAMG55XPLAINED + PL460 (samg55_xplained_pl460)	CEN-A CEN-B FCC	CEN-A CEN-B FCC	CEN-A CEN-B FCC	CEN-A CEN-B FCC
PIC32CXMTG-EK + PL460 + ATREB215 (pic32cxmtg_ek_pl460_xxx_hybrid)	CEN-A CEN-B FCC SUN-FSK SUN-OFDM	CEN-A CEN-B FCC SUN-FSK SUN-OFDM	N/A	N/A

TARGET BOARD (Project folder prefix)	PHY_SNIFFER_TOOL	PHY_TESTER_TOOL	PHY_TX_TEST _CONSOLE	PHY_PLC_AND _GO
PIC32CXMTSH-DB + PL460 + ATREB215 (pic32cxmtsh_db_pl460_xxx_hybrid)	N/A	CEN-A CEN-B FCC SUN-FSK SUN-OFDM	N/A	N/A
PIC32CXMTS-DB + PL460 + ATREB215 (pic32cxmtc_db_pl460_xxx_hybrid)	N/A	CEN-A CEN-B FCC SUN-FSK SUN-OFDM	N/A	N/A
PL360G55CF + ATREB215 (samg55j19_pl360g55cf_ek_hybrid)	CEN-A FCC SUN-FSK SUN-OFDM	CEN-A FCC SUN-FSK SUN-OFDM	N/A	N/A
PL360G55CB + ATREB215 (samg55j19_pl360g55cb_ek_hybrid)	CEN-B SUN-FSK SUN-OFDM	CEN-B SUN-FSK SUN-OFDM	N/A	N/A
PL485-EK + ATREB215 (samg55j19_pl485_ek_cenb_hybrid)	CEN-B SUN-FSK SUN-OFDM	CEN-B SUN-FSK SUN-OFDM	N/A	N/A
SAMG55XPLAINED + PL460 + ATREB215 (samg55_xplained_pl460_xxx_hybrid)	CEN-A CEN-B FCC SUN-FSK SUN-OFDM	CEN-A CEN-B FCC SUN-FSK SUN-OFDM	N/A	N/A

3.3 G3 Examples Support:

TARGET BOARD (Project folder prefix)	ADP_MAC_ SERIALIZE D_APP	DLMS_APP _COORD	DLMS_APP _DEV	METER / MODEM	DEMO_M ETER_G3	MACRT_PLC _AND_GO
PL360MB (sam4cms16c_atpl360mb)	CEN-A CEN-B FCC	CEN-A CEN-B FCC	CEN-A CEN-B FCC	CEN-A CEN-B FCC	N/A	CEN-A CEN-B FCC
PL360BN (same70q21_pl360bn)	CEN-A CEN-B FCC	CEN-A CEN-B FCC	N/A	N/A	N/A	CEN-A CEN-B FCC
PL360G55CB (samg55j19_pl360g55cb_ek)	CEN-B	CEN-B	CEN-B	CEN-B	N/A	CEN-B
PL360G55CF (samg55j19_pl360g55cf_ek)	CEN-A FCC	CEN-A FCC	CEN-A FCC	CEN-A FCC	N/A	CEN-A FCC
PL485-EK (samg55j19_pl485_ek)	CEN-B	CEN-B	CEN-B	N/A	N/A	N/A
PIC32CXMTG-EK + PL460 (pic32cxmtg_ek_pl460)	CEN-A CEN-B FCC	CEN-A CEN-B FCC	CEN-A CEN-B FCC	N/A	N/A	CEN-A CEN-B FCC
PIC32CXMTSH-DB + PL460 (pic32cxmtsh_db_pl460)	CEN-A CEN-B FCC	N/A	CEN-A CEN-B FCC	N/A	CEN-A CEN-B FCC	CEN-A CEN-B FCC
PIC32CXMTC-DB + PL460 (pic32cxmtc_db_pl460)	CEN-A CEN-B FCC	N/A	CEN-A CEN-B FCC	N/A	CEN-A FCC	CEN-A CEN-B FCC
SAM4CMS-DB + PL460 (sam4cms_db_pl460)	CEN-A CEN-B FCC	N/A	CEN-A CEN-B FCC	N/A	N/A	CEN-A CEN-B FCC
SAME70XPLAINED + PL460 (same70_xplained_pl460)	CEN-A CEN-B FCC	CEN-A CEN-B FCC	N/A	N/A	N/A	CEN-A CEN-B FCC
SAMG55XPLAINED + PL460 (samg55_xplained_pl460)	CEN-A CEN-B FCC	CEN-A CEN-B FCC	CEN-A CEN-B FCC	N/A	N/A	CEN-A CEN-B FCC

TARGET BOARD (Project folder prefix)	ADP_MAC_ SERIALIZE D_APP	DLMS_APP _COORD	DLMS_APP _DEV	METER / MODEM	DEMO_M ETER_G3	MACRT_PLC _AND_GO
PIC32CXMTG-EK + PL460 + ATREB215 (pic32cxmtg_ek_pl460_xxx_hybrid)	CEN-A CEN-B FCC SUN-FSK SUN-OFDM	CEN-A CEN-B FCC SUN-FSK SUN-OFDM	CEN-A CEN-B FCC SUN-FSK SUN-OFDM	N/A	N/A	N/A
PIC32CXMTSH-DB + PL460 + ATREB215 (pic32cxmtsh_db_pl460_xxx_hybrid)	CEN-A CEN-B FCC SUN-FSK SUN-OFDM	N/A	CEN-A CEN-B FCC SUN-FSK SUN-OFDM	N/A	CEN-A CEN-B FCC SUN-FSK SUN-OFDM	N/A
PIC32CXMTC-DB + PL460 + ATREB215 (pic32cxmtc_db_pl460_xxx_hybrid)	CEN-A CEN-B FCC SUN-FSK SUN-OFDM	N/A	CEN-A CEN-B FCC SUN-FSK SUN-OFDM	N/A	CEN-A FCC SUN-FSK SUN-OFDM	N/A
PL360G55CF + ATREB215 (samg55j19_pl360g55cf_ek_hybrid)	CEN-A FCC SUN-FSK SUN-OFDM	CEN-A FCC SUN-FSK SUN-OFDM	CEN-A FCC SUN-FSK SUN-OFDM	N/A	N/A	N/A
PL360G55CB + ATREB215 (samg55j19_pl360g55cb_ek_hybrid)	CEN-B SUN-FSK SUN-OFDM	CEN-B SUN-FSK SUN-OFDM	CEN-B SUN-FSK SUN-OFDM	N/A	N/A	N/A
PL485-EK + ATREB215 (samg55j19_pl485_ek_cenb_hybrid)	CEN-B SUN-FSK SUN-OFDM	CEN-B SUN-FSK SUN-OFDM	CEN-B SUN-FSK SUN-OFDM	N/A	N/A	N/A
SAMG55XPLAINED + PL460 + ATREB215 (samg55_xplained_pl460_xxx_hybrid)	CEN-A CEN-B FCC SUN-FSK SUN-OFDM	CEN-A CEN-B FCC SUN-FSK SUN-OFDM	CEN-A CEN-B FCC SUN-FSK SUN-OFDM	N/A	N/A	N/A

4 New Features / Changes

- New G3 Hybrid features supported:
 - Jitter and Cluster Head for RREQ forwarding
 - Trickle for broadcast forwarding
 - Last Gasp functionality
 - Adaptive Tx Power for RF
 - Configurable preamble length for PLC
 - UDP responder extension for Conformance testing
- Changes to G3 Stack:
 - MAC RT redefinition to include more functionalities and provide a standard IEEE-802.15.4 frame Tx/Rx interface
 - ADP and LOADng layers independent from hybrid or single medium configuration, decisions made at execution time, not compilation time
 - LBP module separated from ADP library and provided as source. Integrated some coordinator functionalities in new LBP module
 - Integrate Hybrid Abstraction Layer and MAC Wrapper into a single module
 - Implement AdpNoIPDataRequest function to send ADP frames other than IPv6 ones
- Support for new platforms:
 - PLC:
 - PIC32CXMTG-DB + PL460
 - Hybrid PLC&RF:
 - PIC32CXMTG-EK + PL460 + ATREB215
 - PIC32CXMTSH-DB + PL460 + ATREB215
 - PIC32CXMTG-DB + PL460 + ATREB215
- Obsolete platforms
 - ATPL250AMB
 - ATPL250ABN
- Obsolete projects
 - G3 Gateway

5 Resolved Issues

- PIC32CXMTx platform was not using HW Crypto capabilities.

6 Known Issues

- None.

7 Memory Footprint

7.1 PL360/PL460 Binaries

Size of different binaries, in bytes.

PHY			MAC RT		
CEN-A	CEN-B	FCC	CEN-A	CEN-B	FCC
61191	61530	63297	69550	69593	74390

7.2 G3 Stack and other modules

The following tables show the footprint (in bytes) of the parts that comprise a G3 application, in different configurations. All compilations using IAR 9.20.4:

7.2.1 G3 Device with HW Crypto capabilities

	PLC MAC+PHY	RF MAC+PHY	MAC COMMON	ADP	LOAD NG	CRYPTO	IP	DRIVERS	G3 MNG APP	TOTAL
FLASH	13300	34192	3304	25030	14572	7742	24022	18463	3219	143844
RAM	4702	5735	3753	7653	4024	1248	12459	3052	253	42879

7.2.2 G3 Device with SW Crypto libraries

	PLC MAC+PHY	RF MAC+PHY	MAC COMMON	ADP	LOAD NG	CRYPTO	IP	DRIVERS	G3 MNG APP	TOTAL
FLASH	13300	34192	3304	25030	14572	9784	24022	24315	3219	151738
RAM	4702	5735	3753	7653	4024	10368	12459	7225	253	56172

7.2.3 G3 Coordinator with HW Crypto capabilities for Networks up to 500 Devices

	PLC MAC+ PHY	RF MAC+ PHY	MAC COMMON	ADP	LOAD NG	CRYPTO	IP	DRIVERS	COORD APP	G3 MNG APP	TOTAL
FLASH	13300	34192	3304	25030	14572	7742	24022	18463	948	1295	142868
RAM	4702	5735	8249	13553	8940	1248	12459	3052	8040	189	66167

7.2.4 G3 Coordinator with HW Crypto capabilities for Networks up to 2000 Devices

	PLC MAC+ PHY	RF MAC+ PHY	MAC COMMON	ADP	LOAD NG	CRYPTO	IP	DRIVERS	COORD APP	G3 MNG APP	TOTAL
FLASH	13300	34192	3304	25030	14572	7742	24022	18463	952	1295	142872
RAM	4702	5735	8249	14857	39740	1248	30939	3052	32040	189	140751

7.2.5 G3 Stack Serialized to be managed from an external Host MCU/MPU

	PLC MAC+PHY	RF MAC+PHY	MAC COMMON	ADP	LOADNG	CRYPTO	DRIVERS	SERIALIZATION	TOTAL
FLASH	13300	34192	3304	25030	14572	7742	18463	10224	126827
RAM	4702	5735	8249	13553	8940	1248	6672	8939	58038

NOTE:

All G3 Examples run on top of G3 MAC RT which runs on PL360/PL460, so the corresponding MAC RT binary size must be added to the FLASH footprint of provided examples, as it has to be stored on MCU flash to be downloaded to PL360/PL460 upon startup.

8 Revision History

A	11/2023	First version
B	12/2023	Improve tables readability and add missing projects